

From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Intensive longitudinal research often seeks population-level inference on within-person dynamics without sacrificing idiographic estimation. We introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic systems that combines person-specific dynamic modeling with multivariate meta-analytic synthesis. In Stage 1, person-specific vector autoregressive (VAR) models are fit to obtain each individual's dynamic parameter vector and corresponding sampling covariance matrix. In Stage 2, those parameter vectors are synthesized using a multivariate random-effects meta-analytic SEM, allowing population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation while propagating first-stage uncertainty. In a Monte Carlo simulation, we compare MetaVAR with (a) a one-step Bayesian multilevel VAR benchmark implemented in *Mplus* under default or relatively uninformative priors and (b) an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark. Results indicate that MetaVAR's clearest advantage lies in recovery of between-person heterogeneity and calibration of interval estimates for heterogeneity parameters, whereas the one-step Bayesian benchmark remains competitive for population-average effects. An ecological momentary assessment illustration on negative and positive affect further shows that MetaVAR recovers interpretable average dynamics together with substantial heterogeneity in affective set points, inertia, cross-lagged coupling, and innovation structure.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, structural equation modeling

Intensive longitudinal designs, such as ecological momentary assessment (EMA), experience-sampling, daily diary, ambulatory assessment, and measurement-burst protocols, have made it increasingly feasible to collect dense within-person data in everyday contexts (Bolger & Laurenceau, 2013; Bolger et al., 2003; Csikszentmihalyi & Larson, 1987; Nesselroade, 1991; Shiffman et al., 2008; Sliwinski, 2008; Trull & Ebner-Priemer, 2013). These designs have expanded opportunities to study psychological processes as dynamic systems, in which trajectories evolve over time within persons and individuals differ meaningfully in the parameters governing those trajectories (Asparouhov et al., 2018; Bringmann et al., 2013, 2016; Fisher et al., 2011; Hamaker et al., 2018; Rowland & Wenzel, 2020). This shift reinforces a long-standing methodological point: inferences about within-person dynamics cannot be assumed from between-person variation, and questions about regulation, coupling, and stability require models that distinguish idiographic (person-specific) from nomothetic (population-level) structure (Curran & Bauer, 2011; Fisher et al., 2018;

Gates et al., 2023; Hamaker et al., 2005; Hunter et al., 2024; Molenaar, 2004; Wang et al., 2026).

At the same time, applied researchers rarely want only a collection of person-specific estimates. They also want population-level summaries of average dynamics, between-person heterogeneity, and the ways dynamic features co-vary across people, as well as a basis for relating those person-specific dynamic signatures to covariates and distal outcomes (Beltz et al., 2016; Li et al., 2022). Existing workflows, however, often force a trade-off between two unsatisfactory options. One-step hierarchical models with partial pooling—including multilevel and Bayesian multilevel formulations—can support population inference, but may become computationally demanding and shrink person-specific dynamics toward the population mean. At the other extreme, fully unpooled “fit-many-then-summarize” workflows are modular and scalable, but uncertainty-uncorrected versions often ignore first-stage uncertainty and the multivariate dependence among dynamic parameters, limiting valid inference on the